FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Advanced)-2020

PAPER – 1

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. ABCD

Sol. $X_L = X_C = R$

2. BD

Sol. Stopping potential will not change.

$I \propto \frac{1}{r^2}$

3. CD

Sol. $K = 10\pi = \frac{2\pi}{\lambda}$

$\omega = 50\pi$

4. ABC

Sol. Angular momentum of the planet about the sum is constant.

5. ACD

Sol. When K is open, circuit is balanced wheatstone

$\therefore R_{eq} = \frac{20 \times 40}{20 + 40} + 10 = \frac{70}{3}\Omega$

$\therefore i = \frac{55}{R_{eq}} = 2.36 \text{ A}$

Now when key is closed

We get

$8I_1 + 12(I_1 - I_2) - 40(I' - I_1) = 0 \ .....(i)$

$2I_2 - 12(I_1 - I_2) = -10$ .....(ii)

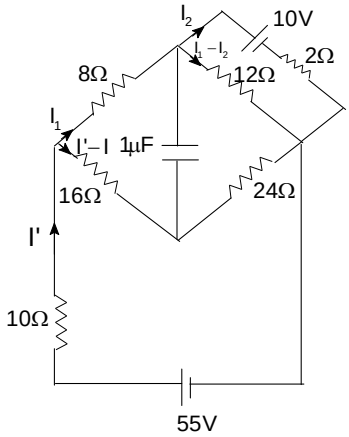
$8I_1 + 12(I_1 - I_2) + 10I' = 55$ .....(iii)

solving (i), (ii) and (iii)

$I_1 = 2A,\ I' = 2.7A,\ I_2 = 1A$

Hence,

Charge on capacitor = 4.8 μc



6. AD

Sol. ∵ Friction is static in nature.

$f = 10N$

Normal contact force = mg

7. ACD

Sol. $B = -\dfrac{dp}{\left(\dfrac{dv}{v}\right)}$ and $PV^2 = constant$

$v^2dp + P(2V)dv = 0$

$vdp = -2pdv$

$\dfrac{vdp}{dv} = -2P$

$\therefore B = 2P$

Similarly we can do for others.

8. D

Sol. If $I = CMR^2$ then

$a = \dfrac{g\sin\theta}{1+C}$

$f = \dfrac{cmg\sin\theta}{1+C}$

$L = \dfrac{1}{2}at^2 \Rightarrow t = \sqrt{\dfrac{2L}{a}}$

and

for solid sphere, $c = \dfrac{2}{5}$

for hollow sphere $c = \dfrac{2}{3}$

for disc sphere $c = \dfrac{1}{2}$

for ring sphere $c = 1$

9. B

Sol. If $I = CMR^2$ then

$$a = \frac{g\sin\theta}{1+C}$$

$$f = \frac{cmg\sin\theta}{1+C}$$

$$L = \frac{1}{2}at^2 \Rightarrow t = \sqrt{\frac{2L}{a}}$$

and

for solid sphere, $c = \frac{2}{5}$

for hollow sphere $c = \frac{2}{3}$

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for ring sphere $c = 1$

10. B

Sol. If $I = CMR^2$ then

$$a = \frac{g\sin\theta}{1+C}$$

$$f = \frac{cmg\sin\theta}{1+C}$$

$$L = \frac{1}{2}at^2 \Rightarrow t = \sqrt{\frac{2L}{a}}$$

and

for solid sphere, $c = \frac{2}{5}$

for hollow sphere $c = \frac{2}{3}$

for disc sphere $c = \frac{1}{2}$

for ring sphere $c = 1$

11. A

Sol. Use Lorentz force

$$\vec{F} = \vec{F}_e + \vec{F}_B$$

$$\vec{F} = q\vec{E} + q\left(\vec{V} \times \vec{B}\right)$$

12. C

Sol. Use Lorentz force

$$\vec{F} = \vec{F}_e + \vec{F}_B$$

$$\vec{F} = q\vec{E} + q\left(\vec{V} \times \vec{B}\right)$$

13. D

Sol. Use Lorentz force

$\vec{F} = \vec{F}_e + \vec{F}_B$

$\vec{F} = q\vec{E} + q\left(\vec{V} \times \vec{B}\right)$

SECTION – B

14. 9

Sol. Field due to the bar magnet at distance x (near the loop)

$$B = \frac{\mu_0}{4\pi}\cdot\frac{2M}{x^3}$$

$\Rightarrow$Flux linked with the loop : $\phi = BA = \pi a^2 \cdot \frac{\mu_0}{4\pi}\cdot\frac{2M}{x^3}$

Emf induced in the loop : $e = -\frac{d\phi}{dt} = \frac{\mu_0}{4\pi}\cdot\frac{6\pi Ma^2}{x^4}\cdot\frac{dx}{dt} = \frac{\mu_0}{4\pi}\cdot\frac{6\pi Ma^2}{x^4}v.$

$\Rightarrow$ Induced current : $i = \frac{e}{R} = \frac{\mu_0}{4\pi}\cdot\frac{6\pi Ma^2}{Rx^4}.v.$

Let F = force opposing the motion of the magnet

Power due to the opposing force = Heat dissipated in the coil per second

$$\Rightarrow Fv = i^2R \Rightarrow F = \frac{i^2R}{v} = \left(\frac{\mu_0}{4\pi}\right)^2 \times \left(\frac{6\pi Ma^2}{Rx^4}\right)^2 \times v^2 \times \frac{R}{v}$$

$$= \frac{9}{4}\left(\frac{\mu_0^2 M^2 a^4 v}{Rx^8}\right) = 36 \times 10^{-22}\ \text{N} \quad \therefore\ y = \mathbf{9}$$

15. 2

Sol. Consider a small element of width dx, radius r situated at a distance x from A. The gradient in radius is uniform. Let it be k.

$$\therefore \quad k = \frac{b-a}{L} \qquad \text{... (i)}$$

If r is the radius of the elementary portion at ef and $(r + dr)$ at gh, then

$$k = \frac{dr}{dx} \text{ ... (ii)}$$

$$\therefore \quad \frac{dr}{dx} = \frac{b-a}{L} \qquad \text{... (iii)}$$

$$dx = \frac{dr}{k}$$

If dl is the extension produced in length dx due to stretching force F using the relation $Y = \frac{\frac{F}{A}}{\frac{\Delta L}{L}}$ we get $\Delta L = \frac{FL}{AY}$

$\therefore \quad dl = \dfrac{F \cdot dx}{\pi r^2 Y}$

$= \dfrac{F \cdot dr}{\pi r^2 Yk}$ since $dx = \dfrac{dr}{k}$

The variable on R.H.S. is r which varies from *a* to *b*.

$\therefore$ Total extension produced $= \int dl = \int_a^b \dfrac{F dr}{\pi r^2 Yk}$

$l = \dfrac{F}{\pi Yk}\int_a^b \dfrac{dr}{r^2} = -\dfrac{F}{\pi Yk}\left[\dfrac{1}{r}\right]_a^b$

$= \dfrac{F}{\pi Yk}\left[\dfrac{1}{a} - \dfrac{1}{b}\right]$

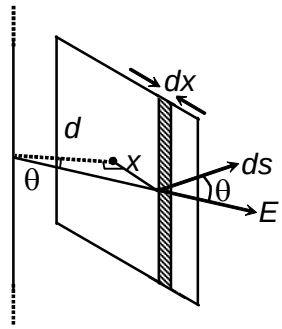
$= \dfrac{F(b-a)}{\pi Ykab}$

But $k = \dfrac{b-a}{L}$

$\therefore \quad l = \dfrac{F.L}{\pi Yab} = 2$ mm

16. 5

Sol. Flux through the element of width *dx* as shown in figure is



$\phi = \int \vec{E}.\vec{ds},\ E = \dfrac{\lambda}{2\pi\varepsilon_0 d\sec\theta}$ and $ds = ldx$

$\phi = \int_{-b/2}^{b/2} \dfrac{\lambda}{2\pi\varepsilon_0}\dfrac{\cos\theta}{d\sec\theta} ldx$

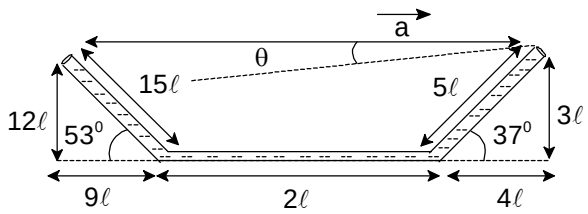
$= \dfrac{\lambda l}{2\pi\varepsilon_0 d}\int_{-b/2}^{b/2}\left(\dfrac{d}{\sqrt{x^2+d^2}}\right)^2 dx$

$= \dfrac{\lambda l}{\pi\varepsilon_0} tan^{-1}\left(\dfrac{b}{2d}\right) = 5$

17. 3

Sol. $\dfrac{9\ell}{15\ell} = \dfrac{3}{5} = \dfrac{a}{g}$

$a = \dfrac{3}{5}g$



18. 4

Sol. $F = idB = (50)\,(0, 1)\,(0, 4) = 2\ N$ …….(1)

As then is no slipping

$F - f = Ma \quad ....(i)$

$fR = I\alpha \quad .....(iii)$

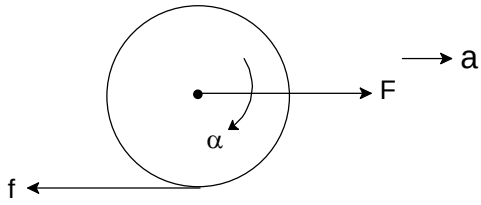
and

$a = \alpha R \quad .....(iv)$

$from(i), (ii), (iii)\ and\ (iv), we\ get$

$a = \frac{2F}{3M} = \frac{4}{9}\ m/s^2$

$\therefore V = \sqrt{2aL} = \frac{4}{3}\ m/s$



Chemistry

PART – II

SECTION – A

19. BD

Sol. The above reaction represent the case of Ar_{SN} (aromatic nucleophilic substitution) which occurs through the formation of benzyne. For the formation of benzyne intermediate, it is essential that an H should be available in on ortho position to the halogen.

20. ACD

Sol. In option A, C, D , less substituted alkene is formed according to Hofmann rule. In option B, more substituted alkene is formed according to Saytzeff rule.

21. AD

Sol. Use Nerst equation, E_{cell} increases when [cathode] increases and that [anode] decreases.
Half cell X is cathode, while Y is anode.

22. ACD

Sol. Average velocity = $\frac{v_1+v_2+v_3+...+v_n}{N}=\sqrt{\frac{8RT}{\pi M}}=\sqrt{\frac{8PV}{\pi M}}=\sqrt{\frac{8P}{\pi d}}$

23. BCD

Sol. $[Cu(NH_3)_4]^{2+}$, $[Pt(NH_3)_2Cl_2]$, $[Ni(CN)_4]^{2-}$ have square planar geometry.

24. AD

Sol. $T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1}$, $w = -dE = -C_V dT = -C_V(T_2 - T_1)$

25. BCD

Sol. $3XeF_4 + 6H_2O \rightarrow 2Xe + XeO_3 + 12HF + 3/2\ O_2$

$XeO_3 + XeOF_4 \rightarrow 2XeO_2F_2$

$XeF_2 + MF_5 \rightarrow [XeF]^+ [MF_6]^-$ (M = As, Sb, P)

$XeF_4 + SbF_5 \rightarrow [XeF_3]^+[SbF_6]^-$

26. B

Sol. factual

27. D

Sol. factual

28. C

Sol. factual

29. C

Sol. factual

30. D

Sol. factual

31. A

Sol. factual

SECTION – B

32. 3

Sol. Ratio of observed and expected osmotic pressure = van't Hoff factor (i) $= \frac{\Delta T_f}{K_f m}$

33. 4

Sol. number of maxima's in radial probability function curve = $n - l$

34. 2

Sol. In Hall – Heroult process, to obtain each kg of aluminium, almost 0.5 kg of graphite is approximately consumed.

35. 4

Sol. $Ag^+ + 2NH_3 \rightleftharpoons [Ag(NH_3)_2^+]$

36. 9

Sol. Number of dipeptides $3^2 = 9$

Mathematics

PART – III

SECTION – A

37. BD

Sol. The equation can be written as

$$\frac{d}{dx}\left(e^{-x}\left(f'(x)-f(x)\right)\right)=2$$

$$\Rightarrow e^{-x}\left(f'(x)-f(x)\right)=2x+c_1$$

$$\Rightarrow f(x)=\left(x^2+c_1x+c_2\right)e^x \text{ and } f'(x)=\left(x^2+(c_1+2)x+c_1+c_2\right)e^x$$

Given that $f'(x)>0 \Rightarrow {c_1}^2-4c_2+4<0$

$$\Rightarrow {c_1}^2-4c_2<0$$

$$\Rightarrow f(x)>0$$

$\therefore$ (B), (D) can be correct.

38. ABCD

Sol. Assume scalar triple product of three vectors

39. ABD

Sol. $|f(0)|\le 0 \Rightarrow f(0)=0$

$$f'(0)=\lim_{n\to 0}\frac{f(n)}{n}$$

Now $\left|\frac{f(n)}{n}\right|\le\frac{\sin^2 n}{n} \Rightarrow \frac{-\sin^2 n}{n}\le\frac{f(n)}{n}\le\frac{\sin^2 n}{n}$

$$0\le\lim_{n\to 0}\frac{f(n)}{n}\le 0 \Rightarrow \lim_{n\to 0}\frac{f(n)}{n}=0$$

$$\Rightarrow f'(0)=0$$

40. BC

Sol. The image of $\bar{a}z+a\bar{z}=0$ with respect to real axis is $\bar{a}\,\bar{z}+az=0$

41. AC

Sol.

$$A=\begin{bmatrix}u_1\\u_2\\u_3\end{bmatrix}\begin{bmatrix}v_1 & v_2 & v_3\end{bmatrix}$$

B C

$$A^5=(BC)^5=BCBCBCBCBC=16A$$

$$B(TrA)^4C=16A$$

$$(TrA)^4A=16A$$

$$\Rightarrow (TrA)^4=16$$

$$\Rightarrow TrA=\pm 2$$

42. BCD

Sol. $x=0: a_0=1$

Differentiations $4\left(1+ax+bx^2\right)^3\left(a+2bx\right)=a_1+2a_2x+....$

Let $x=0 \quad 4a=a_1$

The value of the determinant is $-\left(a_0^3+a_1^3+a_2^3-3a_0a_1a_2\right)=0$

$\Rightarrow \quad \because a_0+a_1+a_2\neq 0$ the $a_0=a_1=a_2=4a$

Differentially again and putting $x=0$

$a=\frac{1}{4}: b=\frac{5}{32}$

43. AC

Sol. Equation of bisectors are given by x – y = 0 and x + y + z = 0

Bisector intersect at the point P(- 1, - 1)

Focus S, is the foot of parallel from P to MN

Where M(1, 1) and N (0, - 2)

MN $\Rightarrow 3x-y-z=0, \quad$ PS $= x+3y+4=0$

$\therefore$ Focus $=\left(\frac{1}{5},\frac{-7}{5}\right)$, MS $=\sqrt{\frac{2}{5}}$, NS $=4\sqrt{\frac{2}{5}}$

LR $=\frac{16\sqrt{2}}{5^{3/2}}$

Circle through PMN is $x^2+y^2-x+y-2=0$

Equation of directrix = equation of tangents at (- 1, -1) is 3x + y + 4 = 0

44. D

Sol. $f(x)=x+\log_e x-x\log_e x$

$f'(x)=1+\frac{1}{x}\log_e x-x\left(\frac{1}{x}\right)=\frac{1}{x}-\log_e x$

$f''(x)=-\frac{1}{x^2}-\frac{1}{x}<0\forall x\in(0,\infty).$

Thus, $f'(x)$ is strictly decreasing function for $x\in(0,\infty)$.

$\left.\begin{aligned}\lim_{x\to\infty}\left(\frac{1}{x}-\log_e x\right)=\lim_{x\to\infty}f'(x)=-\infty\\ \lim_{x\to 0^+}\left(\frac{1}{x}-\log_e x\right)=\lim_{x\to 0^+}f'(x)=\infty\end{aligned}\right\}$

i.e., $f'(x)=$ has only one real root in $(0,\infty)$

$f'(1)=1>0$

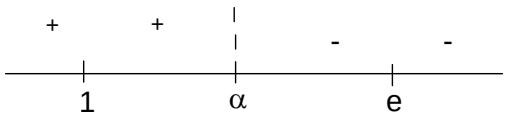
$f'(e)=\frac{1}{e}-1<0$

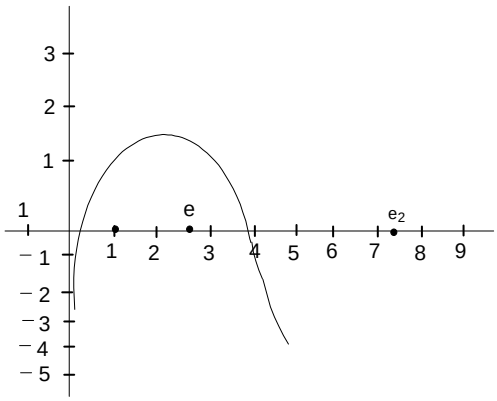
so, $f'(x)=0$ has one root in $(1, e)$.

Let $f'(\alpha)=0$, where $\alpha\in(1, e)$

Therefore, f(x) is increasing in $(0,\alpha)$ and decreasing in (α,∞).

$f(1)=1$ and $f\left(e^2\right)=e^2+2-2e=2-e^2<0$





The graph of the function is as shown in the given figure.

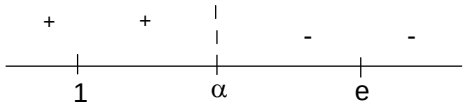
So, f(x) = 0 has one root in $(1, e^2)$

From Column I : I and II are correct.
From Column II : ii, iii and iv are correct.
From Column III : P, Q, S are correct.

45. D

Sol. $f(x) = x + \log_e x - x\log_e x$

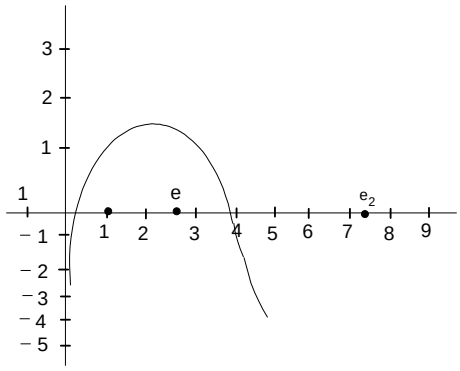
$f'(x) = 1 + \frac{1}{x}\log_e x - x\left(\frac{1}{x}\right) = \frac{1}{x} - \log_e x$



$f''(x) = -\frac{1}{x^2} - \frac{1}{x} < 0 \forall x \in (0, \infty)$.

Thus, $f'(x)$ is strictly decreasing function for $x \in (0,\infty)$.

$\left.\begin{aligned}\lim_{x\to\infty}\left(\frac{1}{x} - \log_e x\right) = \lim_{x\to\infty} f'(x) = -\infty\\ \lim_{x\to 0^+}\left(\frac{1}{x} - \log_e x\right) = \lim_{x\to 0^+} f'(x) = \infty\end{aligned}\right\}$



i.e., $f'(x) =$ has only one real root in $(0, \infty)$

$f'(1) = 1 > 0$

$f'(e) = \frac{1}{e} - 1 < 0$

so, $f'(x) = 0$ has one root in $(1, e)$.

Let $f'(\alpha) = 0$, where $\alpha \in (1, e)$

Therefore, f(x) is increasing in $(0, \alpha)$ and decreasing in (α, ∞).

$f(1) = 1$ and $f(e^2) = e^2 + 2 - 2e = 2 - e^2 < 0$

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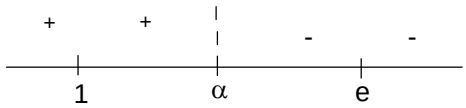
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46. D

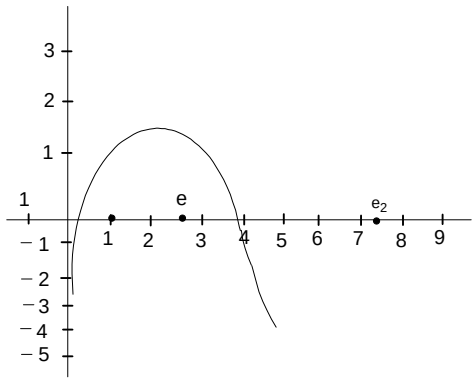
Sol. $f(x) = x + \log_e x - x\log_e x$

$f'(x) = 1 + \frac{1}{x}\log_e x - x\left(\frac{1}{x}\right) = \frac{1}{x} - \log_e x$



$f''(x) = -\frac{1}{x^2} - \frac{1}{x} < 0 \forall x \in (0, \infty)$.

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i.e., $f'(x) =$ has only one real root in $(0, \infty)$

$f'(1)=1>0$

$f'(e)=\frac{1}{e}-1<0$

so, $f'(x)=0$ has one root in $(1, e)$.

Let $f'(\alpha)=0$, where $\alpha \in (1, e)$

Therefore, f(x) is increasing in $(0, \alpha)$ and decreasing in (α, ∞).

$f(1)=1$ and $f(e^2)=e^2+2-2e=2-e^2<0$

The graph of the function is as shown in the given figure.

So, f(x) = 0 has one root in $(1, e^2)$

From Column I : I and II are correct.
From Column II : ii, iii and iv are correct.
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47. C

Sol. $z = x + iy$

$$|z|^2+|z-3|^2+|z-6i|^2$$
$$=x^2+y^2+(x-3)^2+y^2+x^2+(y-6)^2$$
$$=3x^2-6x-12y+3y^2+45$$
$$=3\left[(x-1)^2+(y-2)^2+10\right]$$

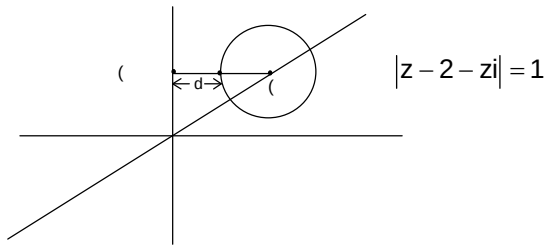
$\therefore$ Minimum value of $\frac{|z|^2+|z-3|^2+|z-6i|^2}{3}=10$

48. B

Sol. $\frac{1}{2}|2iz+4|$

$=|z-2i|$

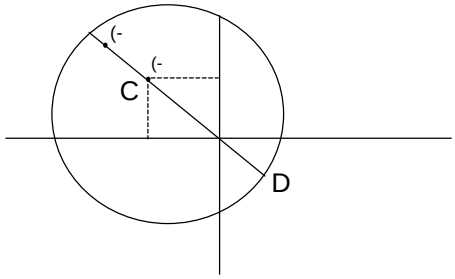
Minimum value of $|z-2i|=d=1$



49. D

Sol. Minimum value of

$$\frac{3}{5}\left|z-\left(-3+\frac{7}{3}i\right)\right|=\frac{3}{5}(CD)=\frac{3}{5}\cdot\frac{20}{3}=4$$



SECTION – B

50. 1

Sol. In hyperbola use property that $x^2/a^2 - y^2/b^2 = 1(a<b)$, if $CP \perp CQ$

Where C is centre then $\frac{1}{CP^2}+\frac{1}{CQ^2}=\frac{1}{a^2}-\frac{1}{b^2}$

51. 2

Sol. $(x^2-y^2)=(2000)^2$

$(x+y)(x-y)=2^2\times10^6=2^8\times5^6$

$(x+y)(x-y)=2^8\times5^6$ (even number)

$\Rightarrow (x+y)$ and $(x-y)$ are even is only possiblity.

no of factors. $7\times7\times2=98$

$\left(\frac{\alpha}{49}=2\right)$

52. 2

Sol. $x^2+y^2+6x-2y+72=0$ .....(i)

$x^2-y^2+6x+16y-46=0$ .....(ii)

(i) + (ii)

$2x^2+12x-8y+26=0$

$x^2+6x-4y+13=0$

$(x+3)^2=4(y-1)$ .....(iii)

pt of intersecting also satisfy equation ...(iii)

$\sum_{i=1}^{4}\sqrt{(xi+3)^2+(yi-2)^2}=\sum_{i=1}^{4}\sqrt{4(yi-1)+(yi-2)^2}=\sum|yi|$

Similarly equation also satisfy

$2y^2-40y+118=0$

$y^2-20y+59=0$

$\Rightarrow \sum_{i=1}^{4} yi = 2\times20=40$

53. 7

Sol. $\left({}^{19}c_0(x^3)^{19}+{}^{19}c_1(px^2+2x-5)(x^3)^{18}+....\right)$

$\left({}^{8}c_0x^{16}+{}^{8}c_1(qx-41)x^{14}+....\right)\left({}^{6}c_0x^{24}+{}^{6}c_1(-x^3+x-1)x^{20}+.....\right)$

$= x^{97} + 391x^{96} + a_{95}x^{95} + ...$

Comparing the coefficient of x^{96} we get

$19p + 8q - 6 = 391$

$\Rightarrow 19p + 8q = 397$ Let $q = 19\lambda + k$, $0 \le k < 19$

$p = \frac{397-8(19\lambda+k)}{19} = 21-8\lambda-2\frac{(4k+1)}{19}$

$\frac{4k+1}{19}$ must be integer $\Rightarrow k = 14$, $p = 15 - 8\lambda$

For minimum positive value of p, $\lambda = 1 \Rightarrow p = 7$

54. 7

Sol. $\sum_{x=1}^{1985} f(x+1) = \sum_{x=1}^{1985} (-1)^{x+1} x - 2\sum_{x=1}^{1985} f(x)$

As f(1) = f(1986)

$\Rightarrow 3\sum_{x=1}^{1985} f(x) = \sum_{x=1}^{1985} (-1)^{x+1} x$

$\Rightarrow \sum_{x=1}^{1985} f(x) = 331.$

Sum of the digits = 3 + 3 + 1 = 7